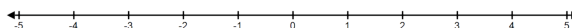


Problem 6

Problem 6

Problem 1 Consider an object moving along a line with velocity $v(t) = \pi \sin(\pi t)$ on $[0, 2]$ and initial position $s(0) = 0$. Time is measured in seconds and velocity in m/s.

- (a) Determine the position function, $s(t)$, on $[0, 2]$.
- (b) Mark the position of the object at the time $t = 1$ on the line below.



- (c) Determine the average velocity, v_{av} , of the object during the interval $[0, 2]$.
- (d) Determine when the motion is in the positive direction.
- (e) At what time (or times) is the object farthest from the origin?